

OptArrow: Interview Findings & Process Recommendations

Date: September 30, 2026

Context: Follow-up interview based on the Evidence-Gap Assessments generated by the AutoDoc V6.4 Agentic Pipeline.

Interviewees: Farid Zare (Core Developer), Zaffar Haider Janjua (Supervisor)

Interviewer: Oluwatimilehin Ayomide Oladipo

1. Executive Summary

The AutoDoc V6.4 pipeline successfully audited the OptArrow repository against ISO/IEC TS 33061:2021. The automated GAP analysis correctly flagged missing evidence regarding requirement traceability, formal architecture governance, and the assessment of technological alternatives.

A developer interview confirmed the accuracy of the V6.4 pipeline: these processes are genuinely absent because OptArrow is currently transitioning from an exploratory academic research project into an established, mature software tool. This document summarizes the human context gathered during the interview and provides lightweight recommendations to achieve full ISO compliance.

2. Interview Findings (Mapped to ISO 33061 Gaps)

Phase 1: Requirements Definition (TEC.3)

- **Gap Identified:** Lack of evidence for continuous feedback review and requirement traceability (BP3, BP4).
- **Interview Finding:** Feedback is gathered informally via GitHub Issues. Users (such as researchers using MATLAB or Julia) can open issues to request features. However, there is no formal agreement process or traceability matrix linking the final codebase back to specific stakeholder requests.

Phase 2: Architecture Definition (TEC.4)

- **Gap Identified:** Missing evidence for architectural roadmapping, candidate assessment, and formal governance (BP1, BP3, BP5, BP6).
- **Interview Finding:**
 - **Evolution:** The architecture evolved organically based on developers' past experiences rather than a formal upfront roadmap.
 - **Rationale:** Apache Arrow was selected for its in-memory data transfer speeds and multilingual compatibility. HiGHS was chosen because it is a popular, free, and open-source solver.
 - **Candidate Assessment:** The team initially used the *Cobra Toolbox* (local model and solver) but assessed that a remote-server architecture was superior for computational load while allowing developers to work on local PCs.
 - **Governance:** There is currently no formal governance or process hierarchy for

approving future architectural changes.

Phase 3: Design Definition (TEC.5)

- **Gap Identified:** Missing evidence of assessing alternative software elements (BP3).
 - **Interview Finding:**
 - **Top-Down Decisions:** No alternatives were evaluated for the IPC transport layer. The Principal Investigator (PI) specifically requested an IPC transport layer, and the students built it using Apache Arrow.
 - **Trial and Error:** For internal process design, alternatives are tested via rapid prototyping. If a feature fails, it is dropped. The only record of these "alternatives" exists in the deleted features and modification history of the GitHub commits.
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3. Process Recommendations for Project Maturity

To transition OptArrow from an academic research project into an established, commercial-grade open-source project (and satisfy ISO 33061 Base Practices), the following lightweight process formalizations are recommended:

1. Implement Traceable GitHub Workflows (Fixes TEC.3):

- *Recommendation:* Enforce a rule that every Pull Request (PR) must be explicitly linked to a GitHub Issue (e.g., `Closes #42`).
- *Benefit:* This creates an automatic, zero-overhead traceability chain from stakeholder request (Issue) to implemented requirement (Code).

2. Adopt Architecture Decision Records (ADRs) (Fixes TEC.4):

- *Recommendation:* Maintain a simple `docs/architecture/decisions/` folder. When a major technology is chosen (like Apache Arrow or HiGHS), write a 1-page markdown document explaining *why* it was chosen and what alternatives were rejected.
- *Benefit:* Provides formal governance and captures the "human rationale" that code alone cannot express.

3. Document Top-Down Constraints (Fixes TEC.5):

- *Recommendation:* When the PI or stakeholders mandate a specific technology (preventing the assessment of alternatives), explicitly document this mandate in the project's design specifications as a "Hard Constraint."
- *Benefit:* Clarifies to future auditors and developers why alternatives were not assessed, fulfilling the requirement for design transparency.